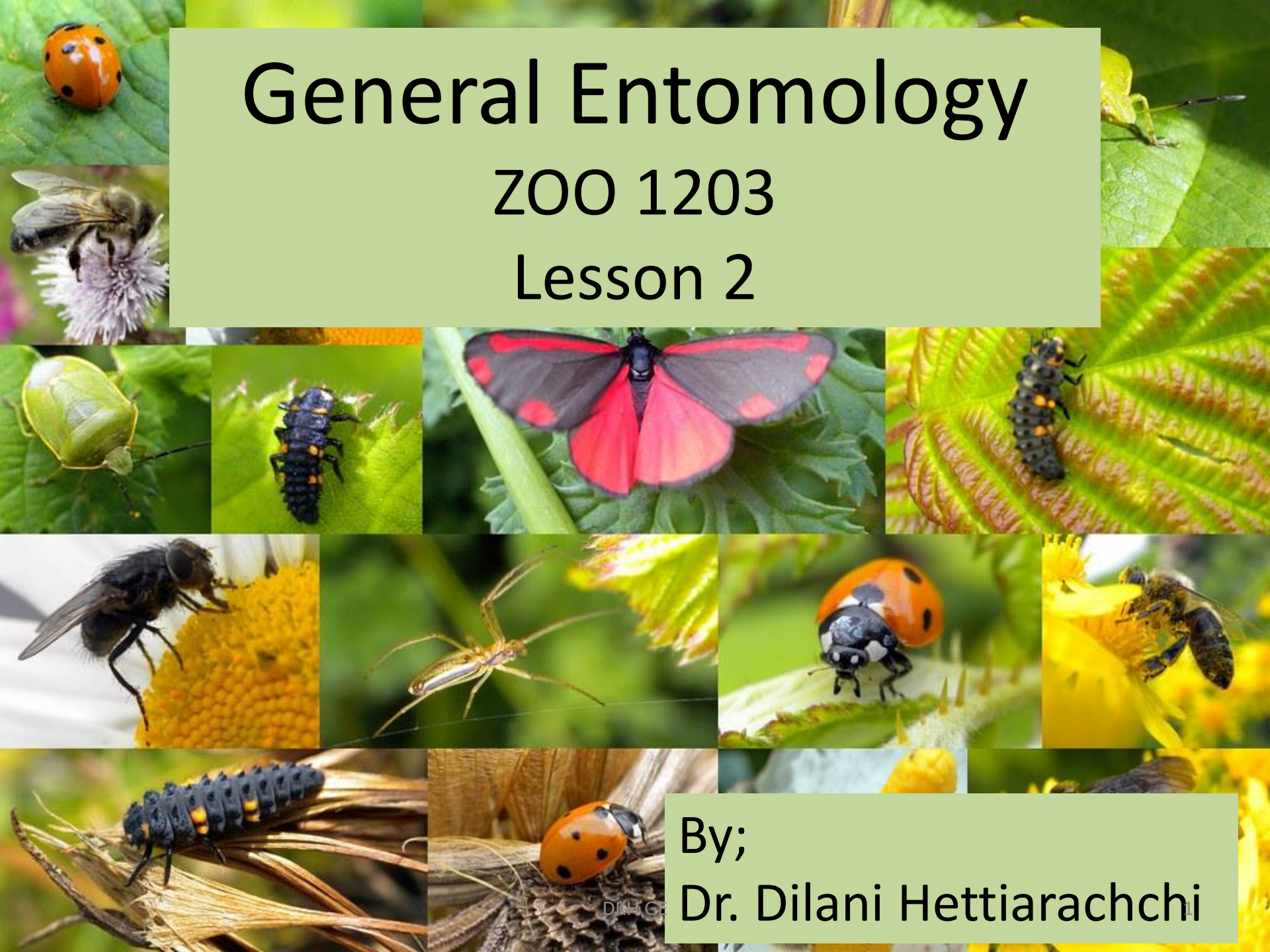


General Entomology

ZOO 1203

Lesson 2



By;

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Lesson ILOs:

At the end of the course students should be able to;

- Describe the characters of insects which helps them to be a dominant species in the world
- Explain why they are considered as one of the dominant species in the world.
- Understand their role in the environment
- Discuss their role as an enemy and a friend of human

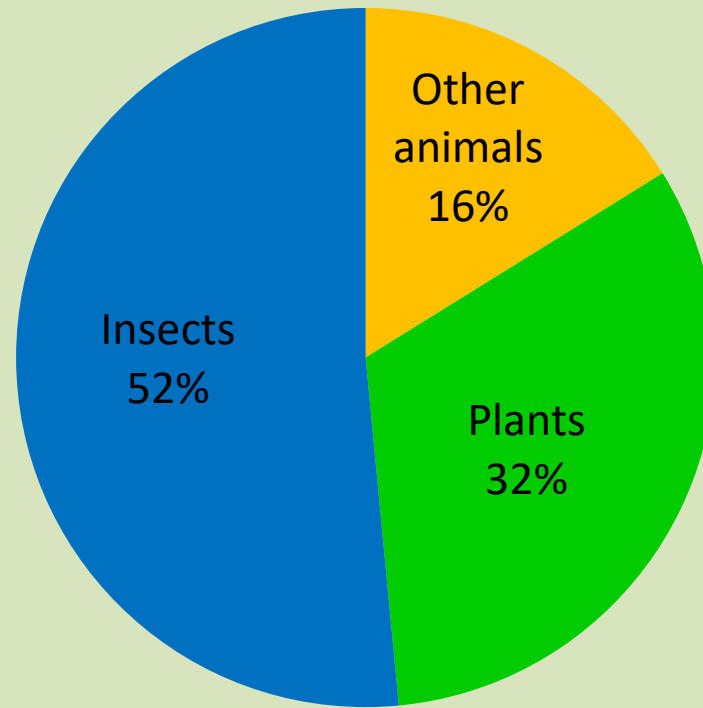


The dominance of Insects

Number of Species

- Comparing the total number of insect species with similar total for other taxonomic groups
- Over 800,000 species have been named and described since Carolus Linnaeus (1707 – 1778)
- Plant kingdom contain between 400,000 – 500,000 species
- Animal kingdom:
 - lower invertebrates – around 200,000 species
 - vertebrates – about 50,000 species

Number of species



- Is the catalogue of life completed?

NO

- What are the chances of discovering a vertebrate?
 - rare event
 - only handful have been found in the past 20 years
- But MILLIONS of undiscovered insect species are still hiding in the soil, in canopies of trees and even in your own backyard
- Entomologists describe hundreds of these new species each year
- still one half to one third of the earth's total insect fauna has yet to be discovered
- Two of every three living species may be insects



Number of individuals

- Another way to look at dominance
- Their small size and high reproductive potential lead them to have high populations
- A swarm of migratory locusts may contain up to 10 billion individuals, cover an area of several thousand hectares, and have a total biomass of over 30,000 metric tons



Distribution

- Third way to assess the dominance
- Insects are found in virtually every terrestrial and fresh-water environment
- From mountain tops to tide pools at the seashore
- In underground caverns (about 12 species)
- Polar ice caps (at least 40 species) and surface of glaciers or near the edge of melting snow fields
- Hot springs, deserts (temp. 60°C)

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- Both above and below the surface of rivers, streams, lakes and ponds
- Only place on earth where insects are NOT plentiful is in the oceans depths



Reasons for success

- Insects have maintain their position for over 400 million years
- They have witnessed:
 - the rise and fall of the dinosaurs
 - survived at least four major extinctions
 - they continued to thrive despite all the obstacles
- For this success insects have unique combination of characteristics

Exoskeleton

- Skeletal is located on the outside of its body
- Gives shape and support to the body's soft tissues
- Provide protection from attack or injury
- Minimize the loss of body fluids in both arid and freshwater environment
- As a “suit of armor” – resist both physical and chemical attack
- Freedom of movement is ensured by membranes and joints in the exoskeleton



Small size

- They range in size from giant moths to tiny parasites
- But most species are between 2 and 20 mm (0.1 – 1 inch) in length
- *Dichomorpha echmepterygis* Costa Rican wasp (family Mymaridae) is the smallest of the small (0.139 mm in length)



- For an animal with an exoskeleton; small body size is important
- If insects are larger as cows or elephants;
 - Exoskeleton would have to be proportionately thicker to support the additional mass of the body
 - If thicker it is heavier and more cumbersome / bulky
 - Even the simplest move would require a larger muscle volume and consume more energy
 - $\text{Length} \times 2 = \text{surface area} \times 4 = \text{volume and mass} \times 8$
 - Thus there is an upper limit to how large insects can become (somewhere around 125 – 150 grams)
 - Beyond this size surface area may not be enough for muscle attachment

- With minimal size →
 - minimal resource is needed for survival and reproduction
 - insect may live on a single plant or animal for its entire life

Eg: A leaf miner, for example, spends its entire larval stage tunneling within the paper-thin layer of cells (parenchyma) between the upper and lower epidermis of a single leaf.



- With minimal size →
 - predators can be avoid easily
 - can hide in the cracks of a rock, beneath the bark of a tree, behind the petal of a flower or under a blade of a grass
 - exoskeleton is hard enough for them to hide yet flexible enough to let them squeeze through the tiniest cracks
- Small size together with adaptations in body shape and coloration gives the ability to blend with their environment (camouflage)
- Thus they are virtually undetectable



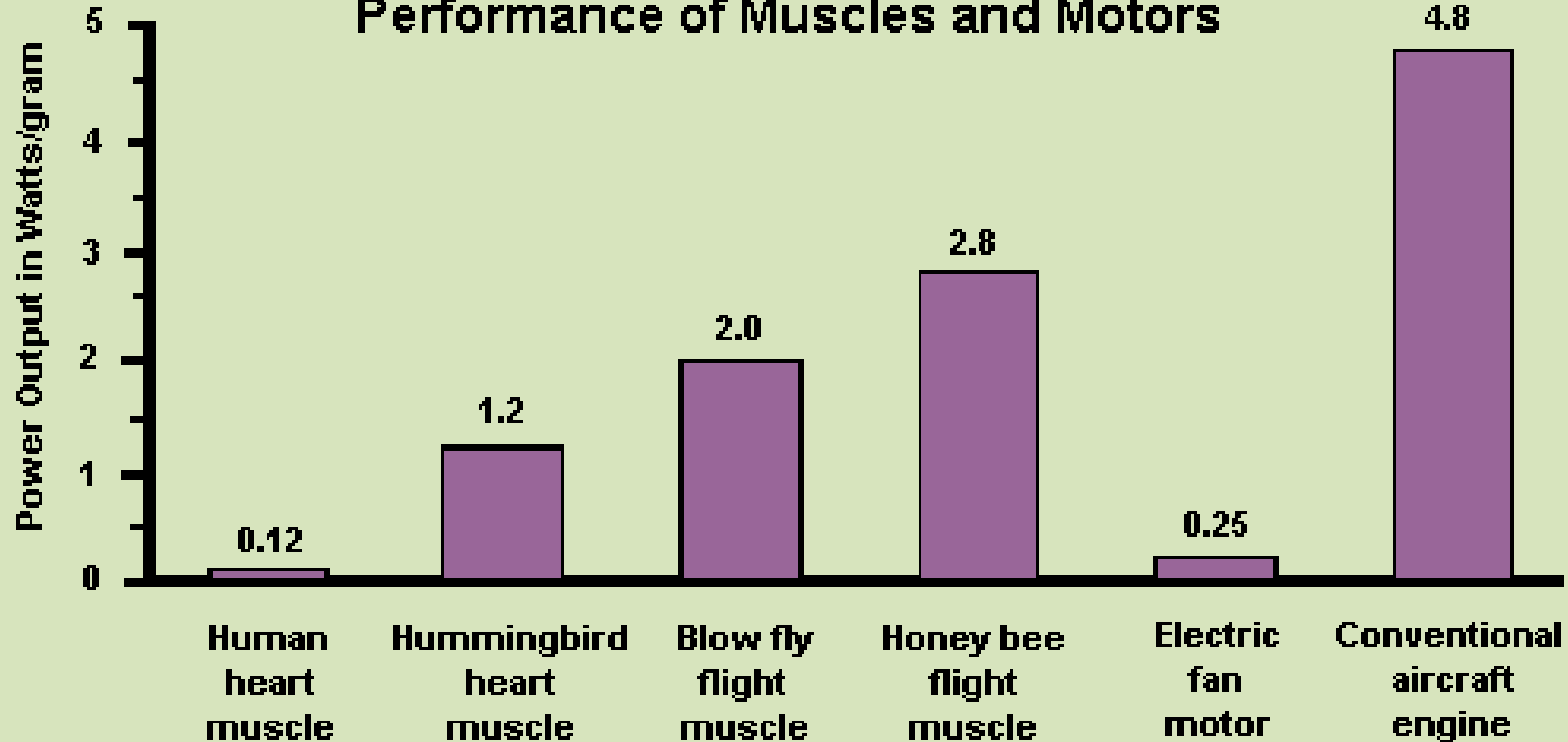
Can you spot the insect?



Flight

- Insects are the only invertebrates that can fly
- It is a highly effective mode of escape from predators
- Efficient means of transportation → allow populations to expand more quickly into new habitats and exploit new resources
- Employ principles of aerodynamics that human engineers do not yet fully understand
- Flight cost is similar to that of birds and bats
- But their flight musculature produce at least 2x more power per unit of muscle mass

Performance of Muscles and Motors

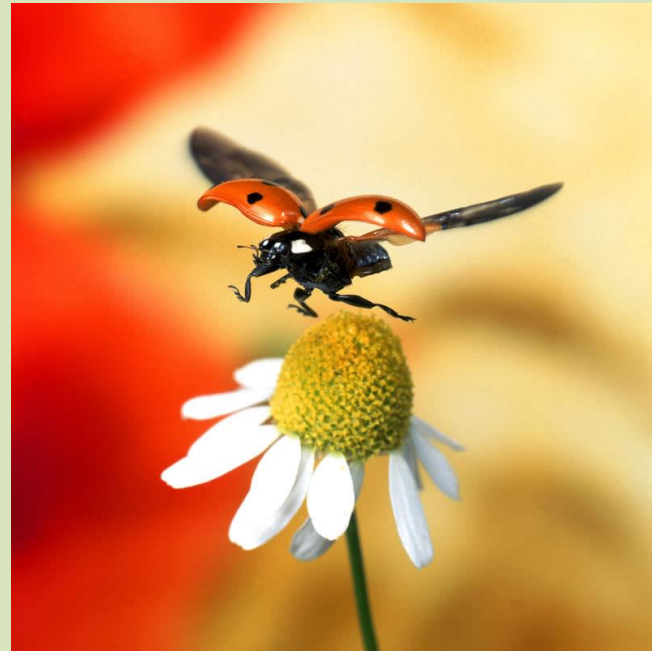


- Some insects travel great distance
- Or remain airborne for a long period of time
- More than 200 species are known to migrate over long distance by air (eg: moths, dragonflies, locusts, flies and beetles)
- Some insects can hover, turn while hovering and even fly backwards
- Some can swift – larger hawk dragonfly can speed up to 58 km/hr; fastest human run only about 36 km/hr



- Some of the insects can carry considerable amount of weight while flying

Eg: Green darner dragonflies (*Anax junius*) able to fly carrying a load up to 15 times their body weight



Reproductive potential

- Most significant measure of an organism's fitness
- Female often produces large numbers of eggs – high fecundity
- Most of the eggs hatch – high fertility
- Life cycle is relatively short – often as little as 2 – 4 weeks
- Together these 3 characteristics enable insects to produce remarkable large numbers of offspring
- A typical female lays 100 – 500 eggs in her lifetime

- Most insects die before they ever have an opportunity to reproduce – one of the reasons for high reproductive potential
- Many adaptations to maximize reproductive potential
 - females can store sperms for months or years within spermatheca
 - a single mating can supply a female with enough sperms to fertile all the eggs she will produce in her lifetime
 - unbalanced sex ratio – more females
 - since most species are not monogamous few males can supply sperm for a large number of females

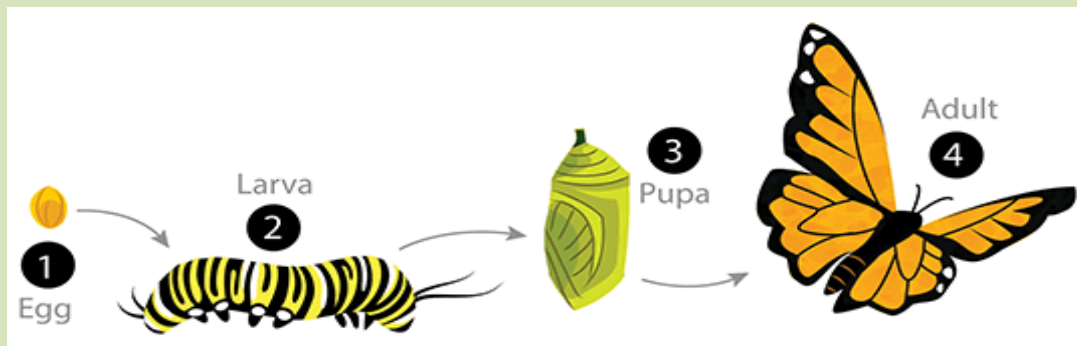
- in many species males are absent (eg: aphids, scale insect, thrips and midges)
- all members of the population are females
contribute offspring through a process of asexual
reproduction



Metamorphosis

- Most insects undergo significant development changes as they grow from immature to adults
- Called metamorphosis
- Involves physical, biochemical, and/or behavioural alterations
- It promote survival, dispersal and reproduction of the species
- In the class Insecta, only 9 out of 28 orders undergo complete metamorphosis - about 86% of all insect species

- Compartmentalization of the life cycle is the advantage
- Larval forms and function is well adapted for growth and development
- Adults are well adapted for dispersal and reproduction
- Each stage is free to adapt to its own ecological role
- i.e. larvae and adult can feed on different types of food and live in different habitats



Adaptability

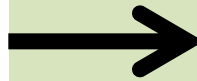
- Large diverse population, high reproductive ability and relatively short life cycle allows insect to quickly adapt to the changing environment by altering/modifying genes
- Their record of achievement is impressive:
 1. First to invade the arid expanses of dry land
 2. First to exploit plant as a source of food
 3. First to use flight as an escape from predators
 4. First to develop a complex social hierarchy with division of labor and corporative care of the young
- For 400 million years they have overcome climatological and geographical disruptions

- For 400 million years they have overcome climatological and geographical disruptions including;
 - evaporation of inland seas
 - formation of mountain ranges
 - shifts in continental plates
 - onset of ice ages
 - fallout from cosmic impacts
- Adaptation is a ongoing process
- Population must continually change as new resource appear and old ones disappear

- Within the few thousand years human began to roam in the world insects starts to consume non-natural things
 - glue and wall paper paste
 - book binding
 - cardboards
 - chocolate
 - cakes
 - pepper
 - stuffed museum specimens, etc...



- Due to this adaptability; insects are capable of achieving resistance to pesticides, insecticides, other chemicals and biological insecticides
- Thus there are instances where this remarkable adaptability has become a problem



Insects as enemies of man

a) Value of insects to man

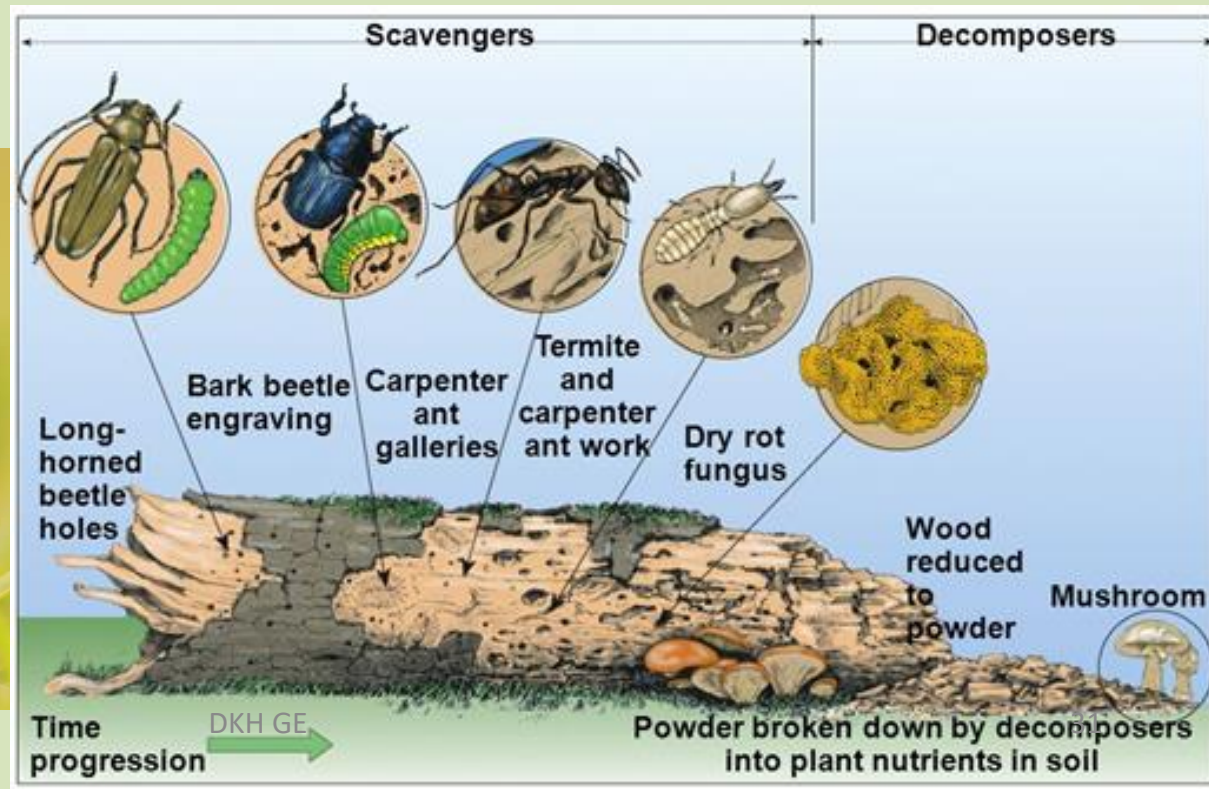
- Plant propagation – pollination and seed dispersal



- Food for insectivorous vertebrates including many birds, mammals, reptiles and fish



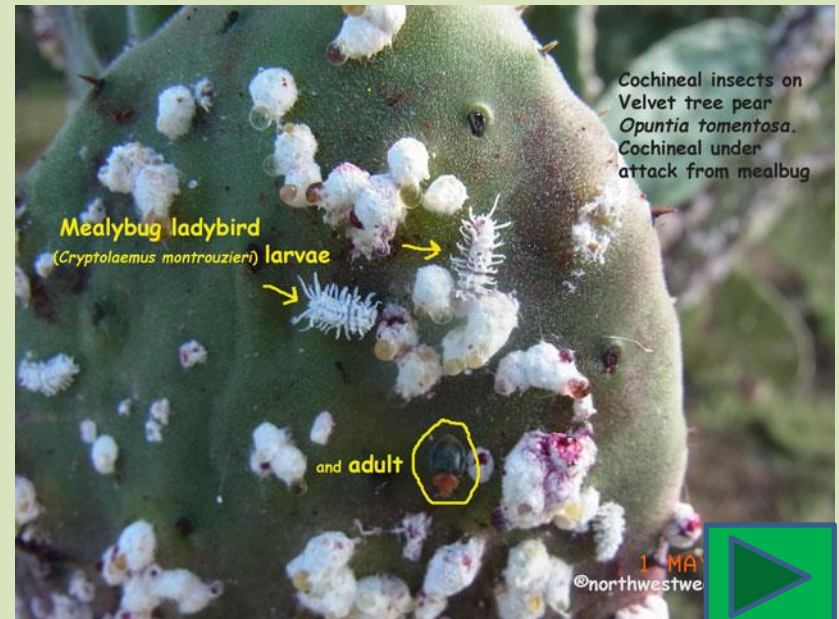
- Nutrient recycling via leaf litter and wood degradation
- Disperse of fungi
- Disposal of dung and carrion
- Soil turnover



- Future food for humans



- Use of secretion of insects (e.g. silk and dye)



- Products from plant galls caused by insects (e.g. tannic acids, ink and dye)



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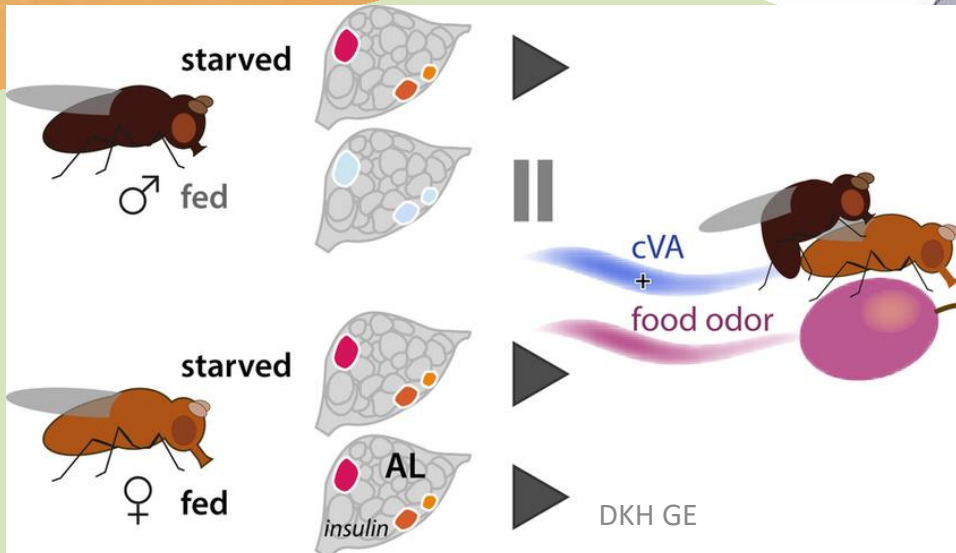
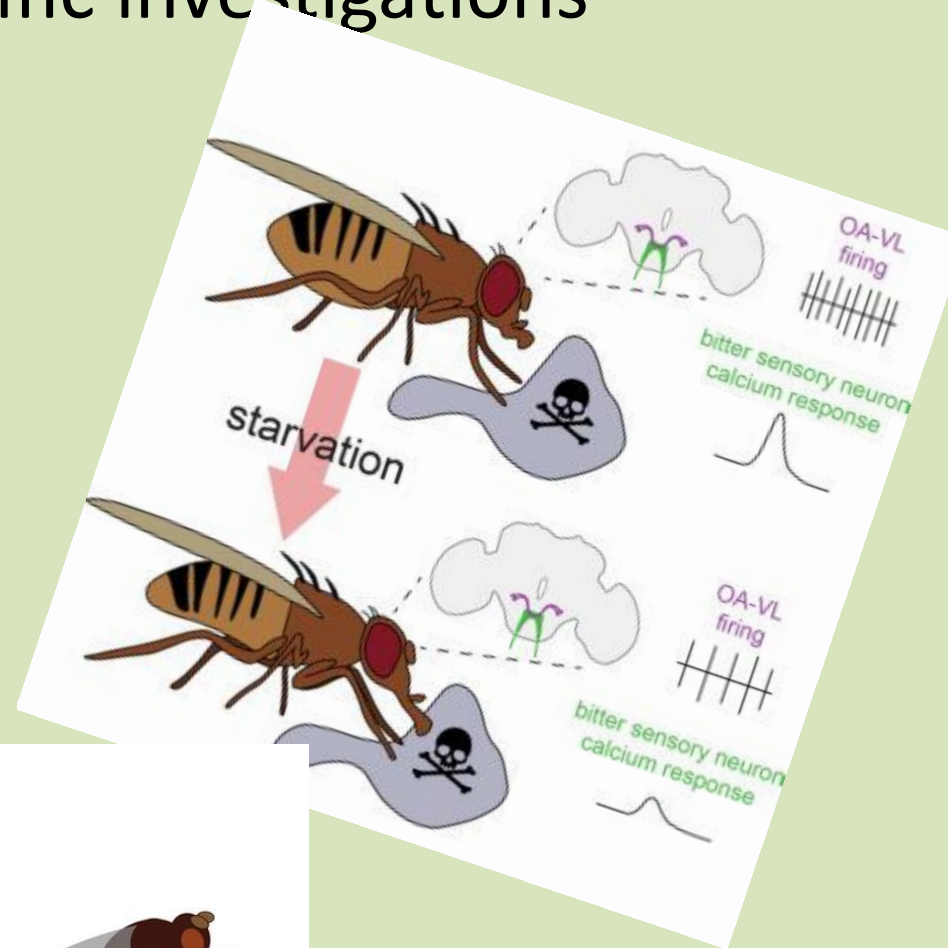


- Predators and parasitoids

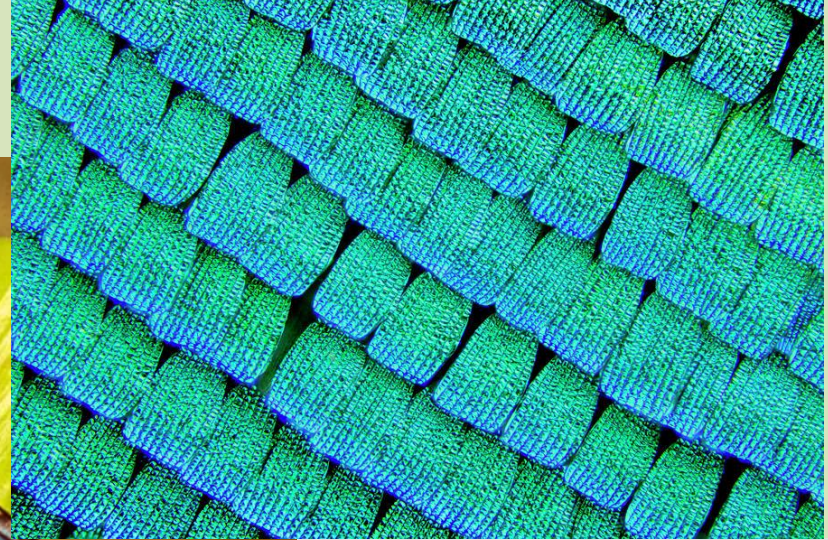


- Ideal material for scientific investigations

Drosophila spp.



- Aesthetic value





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b) Method of injury by insects

- Pests of crop plants



- Pests of stored products



- Household pests
 - Damage by feeding (furniture beetles)



- Contamination by excretion (cockroach)
- Seeking protection/building nests or towers (termites, wasps)



- Insects inimical to man
 - causing annoyance (head louse, bed bugs)
 - causing irritation on body (mosquito, ants, wasps)
 - disseminating diseases (mosquitoes, flea, sand fly, fly)



THANK YOU!



Believe in yourself

